CloakHer (AegisPix)

Adversarial Immunization & Invisible Steganographic Shield for Women's Social Imagery

Competition: Artemia 2.0 (IEEE WIE USJ)	**Theme:** Women's Safety & Cyber Defense	**Aligned Goals:** UN SDGs 5, 9, 11
Track: Ideathon & Designathon	**Platform:** Web & Mobile Application	**Core Tech:** Agentic AI + Adversarial ML

1. Executive Summary

CloakHer is an autonomous protective ecosystem that solves digital sexual harassment, non-consensual deepfake generation, and cyber-blackmail targeting women. Rather than offering a reactive panic button that triggers only after abuse occurs, CloakHer **immunizes personal photographs before they are posted online**. By injecting imperceptible adversarial perturbations and frequency-domain cryptographic marks, the photo remains completely natural to human eyes while rendering itself unprocessable and mathematically disruptive to generative AI models and illicit nudify algorithms.

2. Problem Identification & Prioritization

In accordance with Artemia 2.0 Session 01 guidelines ('*Stay in the problem space — describing your app here is the #1 mistake*'), the core problem is defined strictly around human vulnerability and tangible consequences:

> ***Validated Problem Statement:***
> *Young women and female university students active on social media (**Who**) struggle with the unauthorized scraping and weaponization of their personal photographs into AI-generated non-consensual intimate imagery and blackmail material (**What**) because publicly shared photographs have zero built-in technical defense against generative diffusion models and victims possess no instantaneous method to legally prove synthetic tampering (**Why**), leading to severe psychological trauma, social stigmatization, reputational extortion, academic derailment, and forced withdrawal from digital society (**Consequence**).*

3. Technical Architecture: 3-Layer Defense Engine

To withstand rigorous technical critique, CloakHer abandons fragile visible watermarks (which can be easily cropped or in-painted) in favor of a robust mathematical three-layer defense:

Layer	Methodology	Mechanism & Threat Neutralization
Layer 1: Adversarial Cloaking	Targeted Latent Perturbation (PGD/FGSM)	Injects microscopic changes (±1-2 RGB) across facial contours. When ingested by diffusion models, the noise induces mathematical collapse in the latent space, outputting broken, distorted blurs.

Layer 2: Provenance Signature	DWT-DCT Steganography + C2PA Standard	Embeds tamper-resilient frequency metadata. Survives compression and screenshots. Compliant commercial AI APIs (OpenAI, Midjourney, Adobe) detect the 'Protected Identity' flag and abort generation.
Layer 3: Forensic Verification	Zero-Knowledge Perceptual Hashing (pHash)	Computes one-way perceptual hashes without storing private user photos. If extortion occurs, the user verifies suspicious media and exports an automated, legally formatted Sri Lanka CERT complaint dossier.

4. Agentic AI Integration (Session 02 Alignment)

Directly translating the workshop framework (Conducted by Ravindu Jayaweera) into production capability, CloakHer operates via an autonomous **Plan** → **Act** → **Observe** → **Decide** loop with explicit guardrails:

Step	Agent Role	Technical Operation
1. PLAN	Vulnerability & Landmark Analysis	Scans facial landmarks, skin tones, and background to formulate the optimal minimal perturbation matrix.
2. ACT	Tensor Perturbation & Provenance	Applies adversarial pixel perturbations and injects DWT frequency-domain digital signatures.
3. OBSERVE	Autonomous Self-Evaluation	Attempts to execute an automated internal diffusion reconstruction on the newly cloaked asset to test defense integrity.
4. DECIDE	Convergence & Release	Loops back to calibrate strength if reconstruction succeeds; releases the protected file once immunization exceeds 95%.

Responsible Autonomy & Safety Guardrails (Session 02 Compliance):
• **Zero-Storage Blast Radius Limitation:** Uncloaked photographs are processed exclusively within volatile RAM and deleted immediately after tensor perturbation.
• **Human-in-the-Loop Authority:** The system will never transmit automated complaints to law enforcement or CERT without explicit manual confirmation by the user.

5. Project Scope & Prioritization (Session 01 Standard)

Applying the strict *'Maximum 3 MUST features'* mandate to ensure 100% buildability by Phase 03:

Category	Feature Module	Scope & Deliverable
MUST 1	One-Click Photo Immunization	Upload interface injecting invisible adversarial cloak + invisible watermarking. Instant export.
MUST 2	AI Attack Simulator	Interactive UI component demonstrating side-by-side how an AI generator fails on the cloaked image.
MUST 3	Forensic Tampering Verifier	Upload tool for suspected deepfakes that verifies authenticity and compiles a ready-to-file CERT report.
SHOULD	Social Media Exposure Audit	Guidance scanner evaluating public profile vulnerability to automated scraping.
COULD	Browser Protection Extension	Right-click context menu wrapper for instant desktop protection prior to upload.
NOT YET	Live Video Frame Perturbation	Real-time webcam adversarial injection for streaming platforms (Deferred to future roadmap).

6. Competition Roadmap & Evaluation Edge

• **Phase 01 (Ideathon):** Submission of comprehensive problem statement, architecture diagrams, and value proposition canvas.
• **Phase 02 (Designathon):** High-fidelity Figma prototype showcasing the 'One-Click Cloaking Experience', 'AI Simulator', and 'Forensic Triage Portal'.

• **Phase 03 (Final Live Pitch): The Showstopper Stage Demo:** On stage, the team presents two identical-looking photos of a female teammate. Feeding the unprotected photo into a generative AI tool shows instant face alteration. Feeding the CloakHer-immunized photo causes the exact same AI engine to fail catastrophically and return a glitched blur live in front of the judges.